COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	2026-09-17	**DATE SUBMITTED:**	2026-09-17
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ACTIVITY NO. 8.1 **Install, Configure, and Manage Enterprise Availability Monitoring via Ansible**

1. PROCEDURE OUTPUT

1. Create a playbook that installs Nagios in both Ubuntu and CentOS. Apply the concept of creating roles.
2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)

Ubuntu Playbook:

```
---
- hosts: all
  become: true
  tasks:

    - name: Install Nagios (Ubuntu)
      apt:
        name:
          - nagios4
          - apache2-utils
          - apache2
          - libapache2-mod-php
          - libgd-dev
          - python3-passlib
        state: latest
        update_cache: yes
      when: ansible_distribution == "Ubuntu"
```

- First i made the first 2 common tasks to do in the playbook (hosts and become). And then i made the “Install Nagios” task on Ubuntu. I named the task “Install Nagios (Ubuntu)” to easily determine the task when i ran the playbook. Next I list down all necessary packages/utils for nagios, setting the state to “latest” so that it will install if there’s nothing and update if there’s any. And then update the system via “update_cache” and then lastly i added “when” statement so that it will only do the task solely for Ubuntu only.

CentOS Playbook:

```
    - name: Enable CRB Repository (CentOS)
      ansible.builtin.command: dnf config-manager --set-enable crb
      when: ansible_distribution == "CentOS"

    - name: Install Nagios (CentOS)
      dnf:
        name:
          - nagios
          - httpd
        state: latest
        update_cache: yes
      when: ansible_distribution == "CentOS"
```

- In CentOS, things are different. I need to enable the CRB repo first using the ansible builtin command. What does this do is that it will execute the command “dnf config-manager” directly on the target host.
- Just like the previous task, i installed the necessary packages/utils, update the system, and set it only to CentOS.

3. Show an output of the installed Nagios for both Ubuntu and CentOS.

Ubuntu

Server 1

```
ubuntu1@server1:~$ nagios4 --version

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```

Server 2:

```
ubuntu2@server2:~$ nagios4 --version

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```

CentOS:

```
[centos1@localhost ~]$ nagios --version

Nagios Core 4.4.14
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rs
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```

4. Make sure to create a new repository in GitHub for this activity.
- Github Link: https://github.com/mo-na-min/CPE232_Duenas_nagios-install

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What are the benefits of having an availability monitoring tool?
    - It spots system crashes and sends a message to the IT group right away, so you can handle the problem before people complain.
    - It traces the failure to the real cause, which helps your team stop guessing and restore services sooner.
    - It keeps clear records and gives summary reports, so you can show that you meet your service promises, including SLAs.

Conclusion:

I began by installing Nagios on Ubuntu. Then I used an Ansible playbook to automate the setup. When we moved the work to CentOS, a few syntax issues showed up, so I had to correct them. After that, I went back through the steps and adjusted the tasks. The goal was to keep them from failing on mixed Linux machines. To finish, I made a new github repository to push my tasks onto that.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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